

Practice Question Bank – Data Structures

1. The time complexity of accessing the k^{th} element in an array is _____.
2. The worst case time complexity of linear search is _____.
3. Binary search works only on _____ arrays.
4. In selection sort, the number of swaps in the worst case is _____.
5. The space complexity of merge sort is _____.
6. Which of the following is an in-place sorting algorithm? (a) Merge sort (b) Selection sort (c) Counting sort (d) Radix sort
7. Which time complexity is best for searching? (a) $O(n)$ (b) $O(\log n)$ (c) $O(n \log n)$ (d) $O(n^2)$
8. Which algorithm repeatedly selects the minimum element? (a) Bubble sort (b) Insertion sort (c) Selection sort (d) Quick sort
9. Which case gives the minimum comparisons in linear search? (a) Element at middle (b) Element at last (c) Element at first (d) Element not present
10. Which is not a stable sorting algorithm? (a) Insertion sort (b) Bubble sort (c) Selection sort (d) Merge sort
11. Trace selection sort on [64, 25, 12, 22, 11] for two passes.
12. Trace insertion sort on [9, 5, 1, 4, 3] for two iterations.
13. Show the pivot position after first partition in quick sort for [8, 3, 7, 6, 2].
item Trace binary search for 18 in [2, 6, 10, 14, 18, 22].
14. When does insertion sort perform better than selection sort?
15. Which sorting algorithm would you use for nearly sorted data? Why?
16. Design an algorithm to find the second largest element in an array.
17. Design an algorithm to check whether an array is sorted.
18. Write steps to rotate an array by one position to the right.
19. Design a function to count even numbers in an array.
20. Can a sorting algorithm have best case $O(n)$ and worst case $O(n^2)$? Give an example.
21. Why do we study asymptotic complexity instead of actual time?
22. Can two different algorithms have the same time complexity but different running times? Explain.

23. Predict outputs

(a) For $A = [2, 5, 1, 8, 4]$

```
for (i = 0; i < 5; i++)
{
    if (A[i] < 5)
        print(A[i]);
}
```

(b) For $A = [1, 4, 6, 3, 9]$

```
sum = 0;
for (i = 0; i < 5; i++)
{
    if (A[i] > 3)
        sum = sum + A[i];
}
print(sum);
```

(c) For $A = [10, 5, 8, 1, 2]$

```
for (i = 0; i < 5; i = i + 2)
    print(A[i]);
```

(d) For $A = [3, 6, 1, 7, 4]$

```
count = 0;
for (i = 0; i < 5; i++)
{
    if (A[i] % 3 == 0)
        count++;
}
print(count);
```

24. Complete the codes

(a) Insert an element at the beginning of an array

```
for (i = _____ ; i >= 0 ; i--)
    A[i+1] = A[i];
```

```
A[_____] = x;
n = n + 1;
```

(b) Delete the last element of an array

```
n = _____ ;
```

(c) Insert an element at the end

```
A[____] = x;  
n = _____ ;
```

(d) Delete element at position pos

```
for (i = _____ ; i < _____ ; i++)  
    A[i] = A[i+1];
```

```
n = _____ ;
```

(e) Shift all elements one position to the left

```
for (i = _____ ; i < _____ ; i++)  
    A[i] = A[i+1];
```

25. Define Big- Θ notation. Find the time complexity of finding the maximum element in an array.
26. Define Big- O notation. What is the worst-case time complexity of inserting an element at the end of an array? Justify.
27. Define Big- Ω notation. What is the best-case time complexity of linear search?
28. Draw the graph of n , $n \log n$, and n^2 and arrange them in increasing order of growth.
29. Find the worst-case time complexity of reversing an array.
30. Perform linear search for 11 in [5, 8, 11, 2, 9]. Show comparisons.
31. Perform binary search for 25 in [10, 15, 20, 25, 30, 35]. Show low, high, mid at each step.
32. Perform binary search for 7 in [1, 3, 5, 7, 9, 11, 13]. List the sequence of middle elements examined.
33. Compare best and worst cases of linear search and binary search.
34. Compare linear search and jump search.
35. Compare insertion at beginning vs insertion at end of an array.
36. Compare iterative and recursive binary search.
37. Compare searching in sorted vs unsorted array.
38. Write pseudocode for Selection Sort. Perform the first pass on [29, 10, 14, 37].
39. Write pseudocode for Insertion Sort. Show first iteration on [8, 3, 5, 2].
40. Write pseudocode for Bubble Sort. Show first pass on [4, 1, 3, 2].
41. In insertion sort, find the worst-case number of shifts.

42. In selection sort, how many comparisons occur in the worst case?
43. Write and solve the recurrence for binary search (worst case).
44. Write the recurrence for merge sort (best case) and solve it.
45. Write the recurrence for quick sort when pivot divides into 1 : (n-1).
46. Solve $T(n) = T(n - 1) + 1$.
47. Solve $T(n) = 2T(n/2) + n$.
48. True/False?
 - (a) $n^2 = O(n^3)$
 - (b) $n \log n = O(n)$
 - (c) $2^n = O(n!)$
 - (d) $\log n = O(n)$
 - (e) $n = O(n \log n)$
49. Find time complexity
 - (a)

```
for(i = 1; i <= n; i++)
    for(j = 1; j <= n; j++)
        print(i+j);
```
 - (b)

```
for(i = 1; i <= n; i = i*2)
    print(i);
```
 - (c)

```
for(i = n; i >= 1; i--)
    print(i);
```
 - (d)

```
for(i = 1; i <= n; i++)
    for(j = 1; j <= i; j++)
        print(j);
```
 - (e)

```
for(i = 1; i <= n; i++)
    for(j = 1; j <= n; j = j*2)
        print(i+j);
```
50. Write a recursive function to find factorial of n .
51. Write a recursive function to compute Fibonacci(n).
52. Write a recursive function to find sum of digits of a number.